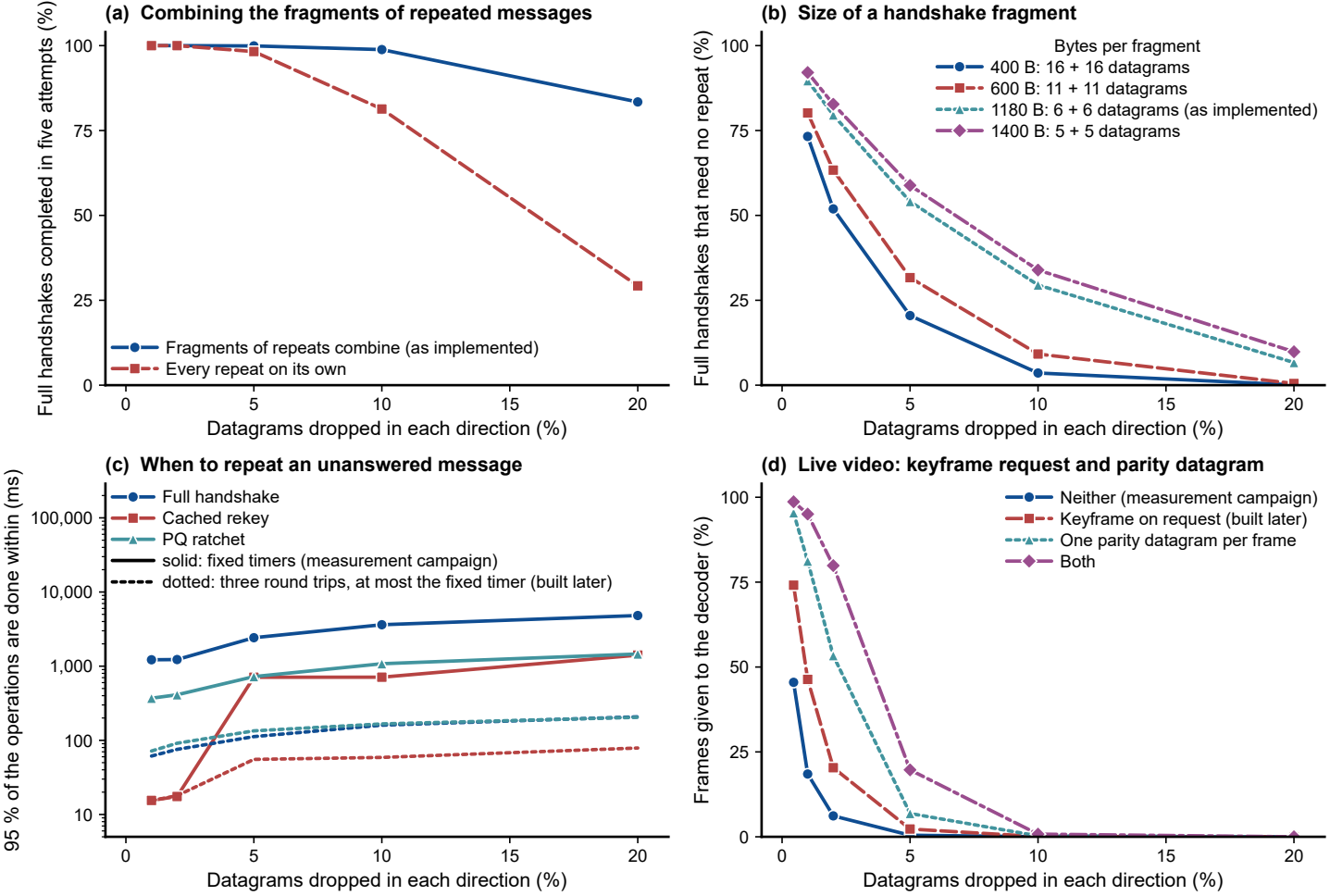
(a) Combining the fragments of repeated messages
Full handshakes completed in five attempts (%)
100
75
50
25
0
Fragments of repeats combine (as implemented)
Every repeat on its own
0
5
10
15
20
Datagrams dropped in each direction (%)
(b) Size of a handshake fragment
Full handshakes that need no repeat (%)
Bytes per fragment
400 B: 16 + 16 datagrams
600 B: 11 + 11 datagrams
1180 B: 6 + 6 datagrams (as implemented)
1400 B: 5 + 5 datagrams
Datagrams dropped in each direction (%)
(c) When to repeat an unanswered message
95 % of the operations are done within (ms)
100,000
10,000
1,000
100
10
Full handshake
Cached rekey
PQ ratchet
solid: fixed timers (measurement campaign)
dotted: three round trips, at most the fixed timer (built later)
Datagrams dropped in each direction (%)
(d) Live video: keyframe request and parity datagram
Frames given to the decoder (%)
Neither (measurement campaign)
Keyframe on request (built later)
One parity datagram per frame
Both
Datagrams dropped in each direction (%)